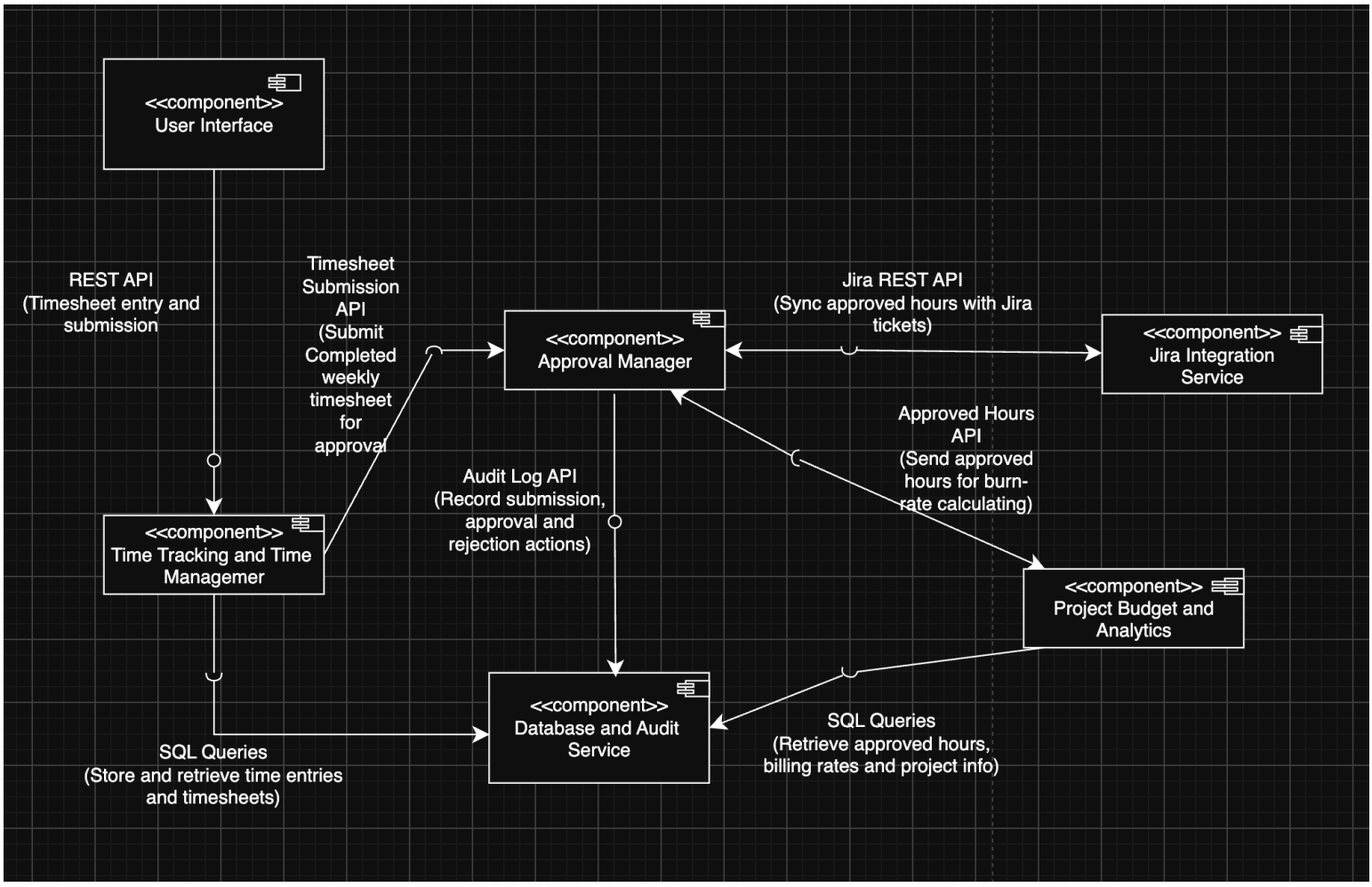
LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Project Title: Remote Team Time-Tracking & Project Approver

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Problem Statement: #46 – Developer Tools & IT Operations

UML Component Diagram

Remote Team Time-Tracking & Project Approver System



ARCHITECTURE ANALYSIS & JUSTIFICATION

1. Architectural Style Analysis

Layered Architecture

- Structure: Divides the system into Presentation, Business Logic, and Data layers.
- Advantages: Clear separation of responsibilities and easy to understand and maintain.
- Disadvantages: May introduce processing overhead and can result in tight coupling between layers.

Microservices Architecture

- Structure: Divides the system into small, independently deployable services.
- Advantages: Independent scaling and better fault isolation.
- Disadvantages: Higher operational complexity, network latency, and data consistency challenges.

Client-Server Architecture

- Structure: Multiple clients communicate with a centralized server.
- Advantages: Centralized control and easier consistency management.
- Disadvantages: The centralized server can become a single point of failure and a scalability bottleneck.

2. Architecture Selection

We chose Layered Architecture for the Remote Team Time-Tracking & Project Approver System.

Then write:

Reason 1 – Separation of Responsibilities:
The system contains distinct responsibilities such as user interaction, time tracking, timesheet approval, Jira integration, budget calculation, and database management.

Layered architecture separates these responsibilities into logical components, making the system easier to understand, maintain, and modify.

Reason 2 – Suitable for Approval Workflow:
The system follows a clear workflow in which developers submit timesheets, managers approve or reject them, approved hours are synchronized with Jira, and project budget burn rates are recalculated. A layered structure supports this sequential flow effectively.

3. Security Advantage

Layered architecture provides controlled access between different parts of the system. Authentication and authorization can be handled before sensitive operations such as timesheet approval, rejection, and audit-log access. This also prevents direct unauthorized access to the database layer.

4. Performance Benefit

The system requires approval status transitions and budget burn-rate information to be updated across dashboards within **500 ms**. Separating business operations from data access allows the system to process requests efficiently and avoids unnecessary direct access to the database from the user interface.